```
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

```
a1=[1,2,3]
type(a)
```

```
numpy.ndarray
```

```
a=np.array([1,3,5,6])
a
```

```
array([1, 3, 5, 6])
```

```
b=[12,13,67]
type(b)
```

```
list
```

```
a2=np.array(b)
```

```
type(a2)
```

```
numpy.ndarray
```

```
c=np.arange(11,29).reshape(3,6)
c
```

```
array([[11, 12, 13, 14, 15, 16],
       [17, 18, 19, 20, 21, 22],
       [23, 24, 25, 26, 27, 28]])
```

```
a=np.random.randint(2,23,5)
a
```

```
array([19, 22, 10, 16, 22])
```

```
q=np.random.randint(2,12 ,size=(5,2))
q
```

```
array([[ 6,  8],
       [ 6,  4],
       [ 8,  7],
       [ 6,  8],
       [11, 10]])
```

```

```

```
a=np.ones(2)
a
```

```
array([1., 1.])
```

```
a=np.ones((2,3))
a
```

```
array([[1., 1., 1.],
       [1., 1., 1.]])
```

```
a=np.zeros(3)
a
```

```
array([0., 0., 0.])
```

```
a=np.zeros((3,3))
a
```

```
array([[0., 0., 0.],
       [0., 0., 0.],
       [0., 0., 0.]])
```

```
a=np.eye(2)
a
```

```
array([[1., 0.],
       [0., 1.]])
```

```
a=np.eye(2,2)
a
```

```
array([[1., 0.],
       [0., 1.]])
```

```
a=np.diag([11,22,33,44])
a
```

```
array([[11,  0,  0,  0],
       [ 0, 22,  0,  0],
       [ 0,  0, 33,  0],
       [ 0,  0,  0, 44]])
```

```
a=np.array([11,22,33])
a
```

```
array([11, 22, 33])
```

```
x=np.diag(a)
x
```

```
array([[11,  0,  0],
       [ 0, 22,  0],
       [ 0,  0, 33]])
```

```
np.random.seed(2)
a=np.random.randint(1,33,4)
a
a.dtype
a=a.astype('float64')
a.dtype
```

```
dtype('float64')
```

```
a=np.random.rand(2)
a
```

```
array([0.54966248, 0.43532239])
```

```
a=np.random.rand(2,2)
a
```

```
array([[0.4203678 , 0.33033482],
       [0.20464863, 0.61927097]])
```

```
a=np.linspace(1,10,8)
a
```

```
array([ 1.        ,  2.28571429,  3.57142857,  4.85714286,  6.14285714,
        7.42857143,  8.71428571, 10.        ])
```

```
a=np.full(2,8)
a
```

```
array([8, 8])
```

```
a=np.array([[12,27],
            [13,56]])
a
a.transpose()
```

```
array([[12, 13],
       [27, 56]])
```

```
a=np.arange(1,4,2)
a
```

```
array([1, 3])
```

```
a**2
```

```
array([1, 9])
```

```
a**3
```

```
array([ 1, 27])
```

```
a=np.arange(1,10).reshape(3,3)
a
```

```
array([[1, 2, 3],
       [4, 5, 6],
       [7, 8, 9]])
```

```
print(np.sum(a))
print(a.sum(axis=1))
print(np.sum(a,axis=0))
```

```
45
[ 6 15 24]
[12 15 18]
```

```
a=np.array([1,2,3,1,1,3,3,4,2,3,4,5])
a
```

```
array([1, 2, 3, 1, 1, 3, 3, 4, 2, 3, 4, 5])
```

```
np.unique(a,return_index=True,return_counts=True)
```

```
(array([1, 2, 3, 4, 5]), array([ 0,  1,  2,  7, 11]), array([3, 2, 4, 2,
1]))
```

```
# pandas
```

```
a={
    'name':['a','c','d'],
    'age':[12,13,14],
    'marks':[22,34,24]
}
```

```
type(a)
```

```
dict
```

```
df=pd.DataFrame(a)
df
```

	name	age	marks
0	a	12	22
1	c	13	34
2	d	14	24

Next steps: Generate code with df | New interactive sheet

```
df.to_csv('prac.csv')
```

```
df1=pd.read_csv('/content/prac.csv')
df1
```

	Unnamed: 0	name	age	marks
0	0	a	12	22
1	1	c	13	34
2	2	d	14	24

Next steps: Generate code with df1 | New interactive sheet

```
df.to_csv('prac1.csv',index=False)
```

```
df2=pd.read_csv('/content/prac1.csv')
df2
```

	name	age	marks
0	a	12	22
1	c	13	34
2	d	14	24

Next steps: Generate code with df2 | New interactive sheet

```
df2.head(2)
```

	name	age	marks
0	a	12	22
1	c	13	34

Next steps: Generate code with df2 | New interactive sheet

```
df2.tail(2)
```

	name	age	marks
1	c	13	34
2	d	14	24

```
df2.shape
```

```
(3, 3)
```

```
df2.size
```

```
9
```

```
df2.sample(3)
```

	name	age	marks
1	c	13	34
0	a	12	22
2	d	14	24

```
df2.columns
```

```
Index(['name', 'age', 'marks'], dtype='object')
```

```
df2.dtypes
```

	0
name	object
age	int64
marks	int64

dtype: object

```
df2.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3 entries, 0 to 2
Data columns (total 3 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   name    3 non-null      object
 1   age     3 non-null      int64
 2   marks   3 non-null      int64
```

```
dtypes: int64(2), object(1)
memory usage: 204.0+ bytes
```

```
df2.describe()
```

	age	marks
count	3.0	3.000000
mean	13.0	26.666667
std	1.0	6.429101
min	12.0	22.000000
25%	12.5	23.000000
50%	13.0	24.000000
75%	13.5	29.000000
max	14.0	34.000000

```
df2_col=df2.select_dtypes(include='object')
df2_col.columns.tolist()
```

```
['name']
```

```
df2.isnull().sum()
```

	0
name	0
age	0
marks	0

dtype: int64

```
df2['name'].value_counts()
```

name	count
a	1
c	1
d	1

dtype: int64

```
df2.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3 entries, 0 to 2
Data columns (total 3 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   name    3 non-null      object
 1   age     3 non-null      int64
 2   marks   3 non-null      int64
dtypes: int64(2), object(1)
memory usage: 204.0+ bytes
```

```
a=df2.iloc[:2,:2]
a
```

	name	age
0	a	12
1	c	13

Next steps: Generate code with a | New interactive sheet

```
b=df2.loc[:2,['name','age']]
b
```

	name	age
0	a	12
1	c	13
2	d	14

Next steps: Generate code with b | New interactive sheet

```
p=df2.dtypes
p
```

	0
name	object
age	int64
marks	int64

dtype: object

```
df2.sort_values(by='age')
```

	name	age	marks
0	a	12	22
1	c	13	34
2	d	14	24

```
df2['age'].mean()
```

```
np.float64(13.0)
```

```
df2.filter(items=['name','age'])
```

	name	age
0	a	12
1	c	13
2	d	14

```
df2[['name','age']]
```

	name	age
0	a	12
1	c	13
2	d	14

```
df2['marks'][:2]=20
```

```
/tmp/ipykernel_5154/3956162841.py:1: FutureWarning: ChainedAssignmentError: b
You are setting values through chained assignment. Currently this works in ce
A typical example is when you are setting values in a column of a DataFrame,

df["col"][row_indexer] = value

Use `df.loc[row_indexer, "col"] = values` instead, to perform the assignment

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/s

  df2['marks'][:2]=20
/tmp/ipykernel_5154/3956162841.py:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/s
  df2['marks'][:2]=20
```

```
df2.head(3)
```

	name	age	marks
0	a	12	20
1	c	13	20
2	d	14	24

Next steps: Generate code with df2 New interactive sheet

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